

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q13-0193

Sw 5Th 138 kV  
December 27, 2014

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0193. The Interconnection Customer proposes a 244.1 MW generating facility interconnecting at the Sw 5Th 138 kV substation in the Pennsylvania area. The request was received on January 05, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 123 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$44,151,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

The LEXINGTON – THREE MILE ISLAND 138 kV is overloaded to 182.8% of its 143 MVA emergency rating for the loss of the SUNBURY 345-138 kV XFMR. The project contributes 22.2% DFAX (54.1 MW) to this overload.

The SW 5TH – THREE MILE ISLAND 138 kV L is overloaded to 134.9% of its 203 MVA emergency rating for the loss of the SUNBURY 345/138kV Autotransformer. The project contributes 22.2% DFAX (54.1 MW) to this overload.

For the loss of SUNBURY 345-138 kV Transformer Bank, the BALBOA – LEXINGTON 138 kV loads to 116.8% of its 208 MVA rating. Project DFAX: 22.2%.

The NILLES CORNER – MAIN ST MO 138 kV is overloaded to 113.8% of its 178 MVA emergency rating for the loss of the ACADIA SUBSTATION – MONTOUR STA 500 kV. The project contributes 7.0% DFAX (17.2 MW) to this overload.

The WEBER CITY – SUNBURY 138 kV is overloaded to 106.9% of its 240 MVA emergency rating for the loss of the ACADIA – MONTOUR STA 500 kV L. The project contributes 7.0% DFAX (17.2 MW) to this overload.

MAUSTON – LONG BEACH 138 kV: 106.4% loading (175 MVA) following the outage of the ACADIA – MONTOUR STA 500 kV L; DFAX = 7.3%.

SW 5TH – ROCK SPRING 138 kV: 105.0% loading (219 MVA) following the outage of the THREE MILE ISLAND – SW 5TH 138 kV; DFAX = 100.0%.

For the loss of SUNBURY 345/138 kV XFMR, the BALBOA – ADLER 138 kV loads to 104.0% of its 222 MVA rating. Project DFAX: 22.2%.

For the loss of ACADIA – MONTOUR STATION 500 kV line, the FONDO DEL SEGURO DEL ESTADO – MAIN ST MO 138 kV loads to 103.9% of its 205 MVA rating. Project DFAX: 7.0%.

WEBER CITY – GREATER DES MOINES ENERGY CTR SUBSTATION 138 kV line: 103.5% loading (225 MVA) following the outage of the ACADIA – MONTOUR STA 500 kV; DFAX = 7.0%.

For the loss of MONTOUR STATION – ACADIA 500 kV, the FONDO DEL SEGURO DEL ESTADO – GREATER DES MOINES ENERGY CTR 138 kV Line loads to 102.7% of its 217 MVA rating. Project DFAX: 7.0%.

FAYETTE – IRONWOOD LLC 345 kV: 101.9% loading (1,244 MVA) following the outage of the CO LINE RD – IRONWOOD LLC 500 kV Line; DFAX = 7.2%.

For the loss of WESTWOOD GENERATION LLC – MORGANVILLE 345 kV L, the UPS CO – WHEELABRATOR FRACKVILLE ENERGY 345 kV loads to 101.0% of its 1,225 MVA rating. Project DFAX: 8.6%.

The WIGWAM – UPS CO SUBSTATION 345 kV L is overloaded to 100.9% of its 1,213 MVA emergency rating for the loss of the MORGANVILLE – WESTWOOD GENERATION LLC 345 kV. The project contributes 9.2% DFAX (22.4 MW) to this overload.

MORGANVILLE – WESTWOOD GENERATION LLC 345 kV: 100.8% loading (1,222 MVA) following the outage of the WIGWAM – UPS CO 345 kV; DFAX = 11.5%.

MAUSTON – BLACK KNIGHT 138 kV: 100.6% loading (200 MVA) following the outage of the ACADIA SUBSTATION – MONTOUR STA 500 kV; DFAX = 7.3%.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-------------------------------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| LEXINGTON – THREE MILE ISLAND 138 kV                                    | Rebuild line with higher-capacity conductor | \$4,204,000   | 100% (\$4,204,000)    |
| THREE MILE ISLAND SUBSTATION – SW 5TH 138 kV                            | Rebuild line with higher-capacity conductor | \$4,923,000   | 100% (\$4,923,000)    |
| BALBOA – LEXINGTON 138 kV                                               | Rebuild line with higher-capacity conductor | \$3,926,000   | 100% (\$3,926,000)    |
| NILLES CORNER – MAIN STREET MO 138 kV Line                              | Rebuild line with higher-capacity conductor | \$14,138,000  | 100% (\$14,138,000)   |
| WEBER CITY – SUNBURY 138 kV                                             | Reconductor line                            | \$1,692,000   | 100% (\$1,692,000)    |
| MAUSTON – LONG BEACH 138 kV                                             | Reconductor line                            | \$4,049,000   | 100% (\$4,049,000)    |
| SW 5TH – ROCK SPRING 138 kV                                             | Reconductor line                            | \$1,306,000   | 100% (\$1,306,000)    |
| BALBOA SUBSTATION – ADLER 138 kV L                                      | Reconductor line                            | \$327,000     | 100% (\$327,000)      |
| FONDO DEL SEGURO DEL ESTADO – MAIN ST MO 138 kV                         | Reconductor line                            | \$1,406,000   | 100% (\$1,406,000)    |
| GREATER DES MOINES ENERGY CTR – WEBER CITY 138 kV Line                  | Reconductor line                            | \$939,000     | 100% (\$939,000)      |
| FONDO DEL SEGURO DEL ESTADO – GREATER DES MOINES ENERGY CTR 138 kV Line | Reconductor line                            | \$1,224,000   | 100% (\$1,224,000)    |
| IRONWOOD LLC SUBSTATION – FAYETTE SUBSTATION 345 kV line                | Reconductor line                            | \$9,833,000   | 9% (\$871,000)        |
| UPS CO – WHEELABRATOR FRACKVILLE ENERGY 345 kV                          | Reconductor line                            | \$8,772,000   | 7% (\$618,000)        |
| WIGWAM – UPS CO SUBSTATION 345 kV L                                     | Reconductor line                            | \$21,524,000  | 7% (\$1,557,000)      |

| Facility                                        | Required Upgrade           | Cost Estimate | Allocation to Project |
|-------------------------------------------------|----------------------------|---------------|-----------------------|
| MORGANVILLE – WESTWOOD<br>GENERATION LLC 345 kV | Reconductor line           | \$9,864,000   | 12% (\$1,193,000)     |
| MAUSTON – BLACK KNIGHT 138 kV                   | Reconductor line           | \$1,778,000   | 100% (\$1,778,000)    |
|                                                 | Total allocated to project |               | \$44,151,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.